

AG-D3QN: An A*-Guided Dueling Double DQN for Efficient AGV Path Planning in Workshops

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Abstract—In order to improve the search efficiency, dynamic adaptability and response speed of workshop AGV (Automated Guided Vehicle) path planning, this study proposes the AG-D3QN (A*-guided Dueling Double Deep Q-Network) algorithm for a single AGV. AG-D3QN cleverly combines the advantages of the A* algorithm in quickly locating the optimal path region with the powerful environment perception and dynamic decision-making capability of D3QN. Experimentally, the algorithm ensures that the training time is significantly lower than that of traditional D3QN while ensuring the efficiency of shortest path finding; its response time can be reduced by 32% compared with the A* algorithm in the same workshop environment, and it shows excellent adaptability and practical value.

Keywords—A*, D3QN, Path Planning Algorithm, AGV

I. INTRODUCTION

The in-depth development of Industry 4.0 is strongly driving the digital and intelligent transformation of the manufacturing industry. As the core platform of the fourth industrial revolution, the digital factory is reshaping the traditional production mode by integrating the Internet of Things (IoT), artificial intelligence (AI) and big data technologies [1]. In this wave of transformation, the demand for upgrading the intelligent logistics system, which assumes the function of ‘artery of production’, is becoming more and more prominent. As the key execution unit of the system, the automatic guided vehicle (AGV) is essential to improve the efficiency of the material flow in the workshop, and its technological evolution and the process of industrial intelligence show a deep synergistic evolutionary relationship.

Accordingly, the global AGV market is undergoing rapid expansion, which not only highlights the urgency of manufacturing intelligence upgrading, but also reveals the core contradiction in the application of the technology: despite the fact that AGVs significantly reduce manpower costs and improve operational safety, their effectiveness is highly limited by the performance of path planning algorithms. This technical bottleneck has become a key factor in maximising the efficiency of AGV systems.

In response to the optimisation needs of path planning algorithms, in recent years, scholars have continuously improved the traditional algorithms. Reference [2] developed a distributed scheduling algorithm based on successive one-way auctions, which effectively realised conflict-free task allocation

under path constraints. Reference [3], by constructing a bi-directional AGV area control architecture and integrating supervisory control theory, successfully eliminated the Reference [4] adopts a time-enhanced A-inspired search strategy to generate feasible paths, but it needs to rely on multi-module collaboration to guarantee the validity of the algorithm's information. Reference [5] extends the search neighbourhood of the traditional A-algorithm to 20-neighbourhoods and introduces a safe-distance mechanism, which significantly enhances the path smoothing degree; however, the improvement leads to a significant decrease in the time consuming and the number of nodes to traverse, and a significant decrease in the algorithm efficiency. However, the improvement leads to a surge in planning time and the number of traversed nodes, and the efficiency of the algorithm decreases significantly. It is worth noting that the inherent heuristic global node expansion mechanism of the A* algorithm is prone to exponential computational complexity due to the redundant search of symmetric paths in shelf-intensive scenarios.

Deep Reinforcement Learning (DRL) provides innovative ideas to solve the problem. It integrates the perceptual advantage of deep learning and the decision-making ability of reinforcement learning to efficiently generate near-optimal paths. In discrete small-scale scenarios, Reference [6] combines an adaptive modal algorithm with Q-learning to deal with the AGV scheduling problem, which performs well in a small-scale state space. Reference [7] draws on the flower pollination algorithm to improve the Q-learning initialisation process, and simulations show that the method is effective in accelerating convergence. Reference [8] improving the convergence efficiency of the scheduling algorithm by improving the action step size of the PPO algorithm performs well in theory. The above results confirm the broad prospect of reinforcement learning in AGV path planning, through autonomous environment interaction and strategy optimisation, AGVs are able to achieve highly adaptive path decisions in complex scenarios.

However, existing methods have difficulty in reconciling dynamic environment adaptability with spatial search efficiency: traditional A* faces the risk of computational explosion, while pure reinforcement learning schemes converge slowly. To break this contradiction, this paper proposes the AG-D3QN algorithm - an efficient and robust AGV intelligent planning by fusing the fast path spatial localisation capability of

A* with the dynamic decision-making advantage of Dueling DQN.

II. ENVIRONMENTAL MODELLING AND RELATED ALGORITHMS

A. State Space

The quality of the state space representation directly determines the performance boundary of the path planning algorithm. In this study, a digital workshop simulation platform is used to provide AGVs with global environment sensing capabilities, covering static topologies such as machining tables, material racks and track layouts, as well as real-time AGV postures and target coordinates. In order to improve the efficiency of the algorithm, a digital coding mechanism for environment information is proposed: the environment is analysed into a state matrix (binary coding: 0 - forbidden zone/1

- feasible zone) and a separated position matrix (independently characterising the current position and target position). The state matrix is dynamically updated according to the AGV operation state: only the current position and the target point are kept as feasible in the material rack area under the loading condition; the non-target points in the processing table area are set as forbidden in the unloaded condition. The position matrix imposes a special constraint on the processing table area - the non-target points are permanently coded as 0 (see Fig. 1 for the process). This mechanism achieves efficient digital mapping of the workshop environment through structured matrices, which has the advantage of accurately capturing the dynamic evolution of the environment, providing low redundancy and strongly interpretable state inputs, and significantly improving algorithmic processing efficiency through dimensional compression.

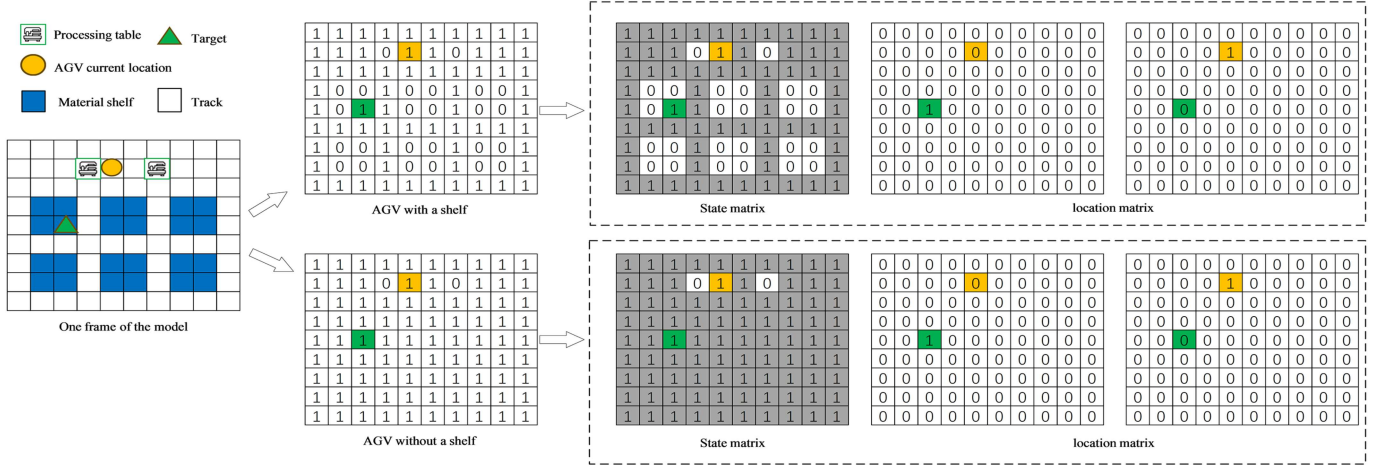


Fig. 1. State Space Single Frame Transition

B. Action Space

In an intelligent manufacturing workshop, the motion control of an AGV is modelled as a 2D planar four-degree-of-freedom action space (top right, bottom left). To accommodate neural network processing, each motion direction is digitally characterised by a discrete vector mapping (see Fig. 2 for the action coding mapping), which is used only as an action identifier and has no actual physical dimensional meaning. When the system is running, the position update of the AGV triggers the transmission of the environment state to the Python backend. The neural network receives the state inputs and outputs action decisions, which are arbitrated in real-time by the action legality verification module: legal actions are executed immediately and the system is maintained, while illegal actions trigger a system safety interruption.

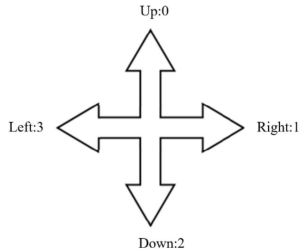


Fig. 2. Action Mapping

C. Reward Structure

The reward function establishes a quantitative evaluation system for AGV decision-making behaviours, and drives the optimization of path planning strategies through the value feedback of state-action pairs. The function is dynamically computed based on the state changes after action execution: positive incentives are given for convergence to the target, and negative penalties are imposed for path deviation or collision events, so that the AGV achieves optimal path exploration in dynamic environments through reinforcement learning. The action space is divided into two categories according to legality: illegal actions (triggering system interruptions and giving -1 penalties): a) out-of-bounds movement b) loaded state hitting material racks c) entering non-target processing table. Legal action: positive action (approaching the target point/completing the stage task: +1 reward) negative action (moving away from the target point: no interruption penalty), and at the same time, introduce the time penalty mechanism (unit displacement deducted 0.01 reward) to improve the efficiency of task execution. The complete reward function is defined in (1).

$$R = \begin{cases} -1 & \text{illegal action} \\ +1 & \text{subtask completed} \\ -0.01 & \text{penalty per step} \end{cases} \quad (1)$$

D. D3QN Algorithm

D3QN algorithm is a high-level architecture for deep reinforcement learning, which optimises dynamic decision-making through dual innovations. Firstly, Dueling Network Separation Mechanism: the traditional Q-network is split into a parallel dual-channel - the state value assessment layer (to measure the potential gain of the environment) and the action advantage assessment layer (to quantify the relative gain of a specific action), which collaborate to output the precise action value. The two work together to output accurate action values, which significantly improves the learning efficiency in sparse reward scenarios. Second, Double Q-learning bias correction mechanism: introducing a dual model of main/target network, effectively suppressing Q overestimation bias by decoupling the action selection and the value assessment (the main network decides the optimal action, and the target network calculates the corresponding value). Combined with the hysteresis update strategy of the target network, the dual Q-learning model can guarantee the training stability and convergence speed, so that it can show strong adaptability in complex scenarios such as AGV path planning, which needs to respond to dynamic obstacles in real time.

In this study, three layers of convolutional neural network hidden layer number of 32, 64, 128, respectively. each layer of convolutional neural network are used 3×3 convolutional kernel, the step size is 1, the padding padding is 1, and the activation function is ReLU, the specific process, as shown in Fig. 3.

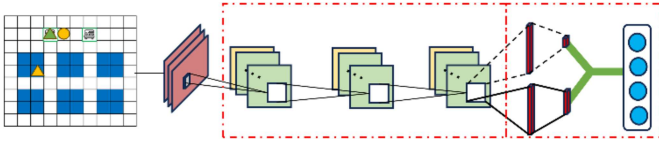


Fig. 3. D3QN network architecture

E. AG-D3QN Algorithm

Aiming at the pain point of low exploration efficiency of D3QN in warehouse AGV path planning, this study innovatively integrates A algorithm to construct a goal-oriented cooperative mechanism. Traditional stochastic exploration suffers from two defects: 1) disordered movement triggers goal deviation and invalid displacement; 2) high probability of generating illegal actions such as loading and hitting shelves leads to system interruption (Fig. 4 shows that only 25% of the actions in picking up goods are positively rewarded, and 50% are triggered by illegality). This scheme regulates the decision weights through the dynamic path sequence: set the attenuation parameter ε to control the neural network and the proportion of A* instructions (initial $\varepsilon = 0.8$, i.e., 20% of A* participation), and adopt the gradual attenuation strategy to make the system transition smoothly from the strong guidance (high A* participation) to the autonomy of decision-making ($\varepsilon \rightarrow 1$), which not only guarantees the convergence of paths at the early stage of training, but also facilitates the network to master the strategy gradually, and ultimately achieves the global optimization balance of the exploration-exploitation. global optimisation equilibrium.

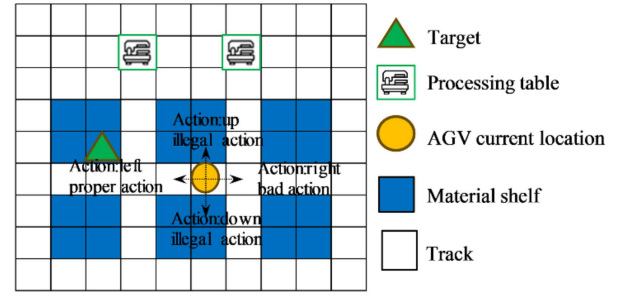


Fig. 4. Randomised Exploration Example

III. EXPERIMENTS

The parameters of the experimental model in this study are shown in Table I, Fig. 5, and the hyperparameters refer to the research results of Reference [9]. The specific parameters are shown in Table II.

TABLE I. MODEL LAYOUT DETAILS

No	Number of stations	Size	Type
Layout1	48	13×12	Rectangular layout
Layout2	48	13×12	Fishbone layout

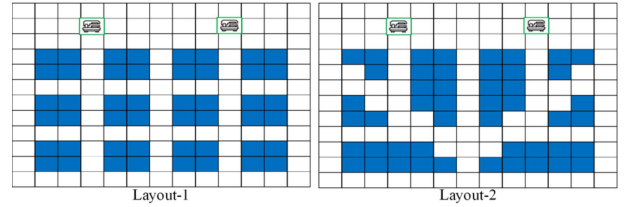


Fig. 5. Model Layout

TABLE II. HYPERPARAMETERISATION

No	Hyperparameter	Value	Role
1	Buffer	20,000	Max experiences stored in replay memory for training.
2	Star_Number	128	Minimum experiences in buffer before training begins.
3	Batch_Size	32	Number of experiences sampled per training step.
4	Gamma	0.95	Discount factor
5	Tau	0.05	Ratio for updating target network weights
6	Learning_Rate	Changing	Step size for neural network optimization
7	Exploring_Rate	Changing	Probability of taking neural network or using A*.

A. Comparison of standard D3QN and AG-D3QN training process

In order to evaluate the robustness of the algorithm under different spatial configurations, a comparative analysis framework between Layout-1 and Layout-2 is designed in this study. In the experiments, D3QN and AG-D3QN are introduced for performance evaluation, and in order to enhance the robustness of the algorithms, the task sequences are randomly assigned in each round in this study. Fig. 6 compares the strategy optimisation process of the two algorithms on two layouts (horizontal axis: number of training iterations; vertical

axis: cumulative reward for a single iteration). The results show that AG-D3QN (green trajectory) reaches the task target reward stably in both layouts. In contrast, D3QN (black trajectory) has a bottleneck in its performance: even after 3,000 iterations, its peak cumulative rewards in Layout-1 and Layout-2 are only around 6 and 11, respectively.

B. AG-D3QN vs A* Response Time Comparison

In order to quantitatively assess the performance advantages of the AG-D3QN algorithm, Table III presents the results of the comparative analysis of the response times of AG-D3QN and the benchmark algorithm A* under two environment

configurations, Layout-1 and Layout-2. The comparison is based on five independent sets of randomly sampled data. We adopt the relative time-saving rate defined in Eq. (2)

$$time_{save} = \frac{time_{A^*} - time_{AG}}{time_{A^*}} \times 100\% \quad (2)$$

For the sake of observation, the important data in Table III is shown in the form of histograms as shown in Fig. 7, Fig. 8. As can be seen from the graphs, AG-D3QN performance is significantly due to the benchmark A* algorithm with less response fluctuation and more stable performance.

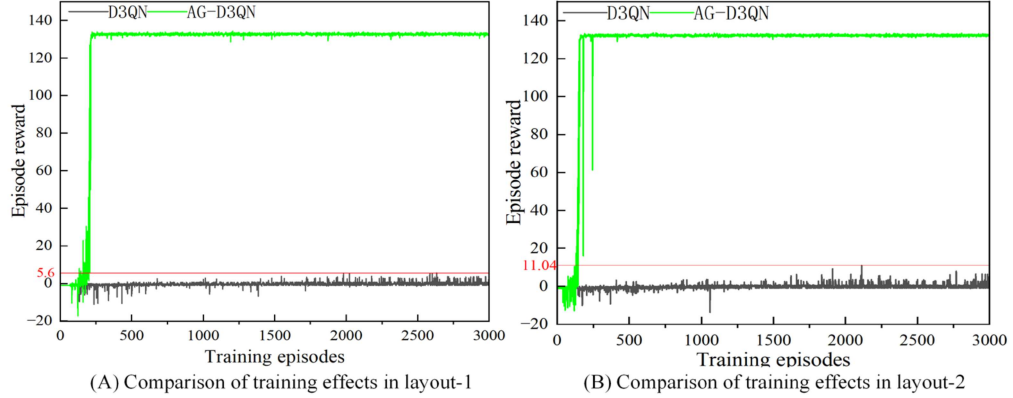


Fig. 6. Training Process

TABLE III. EXPERIMENTAL DATA

Layout	algorithm	Exp1		Exp2		Exp3		Exp4		Exp5		Average response time(s)	Standard Deviation(s)	Saved time
		Response time(s)	Total steps	Response time(s)	Total steps	Response time(s)	Total steps	Response time(s)	Total steps	Response time(s)	Total steps			
Layout1	AG-D3QN	1.2011	1319	1.1251	1273	1.1943	1310	1.0522	1238	1.0473	1261	1.124	0.0662	33.33%
	A*	1.7932	1319	1.6634	1273	1.8497	1310	1.5601	1238	1.5638	1261	1.686	0.1180	
Layout2	AG-D3QN	1.1084	1287	1.1755	1316	1.0272	1198	1.0221	1249	1.1032	1335	1.0872	0.0572	32.09%
	A*	1.6889	1287	1.7615	1316	1.3862	1198	1.4627	1249	1.7063	1335	1.6011	0.1482	

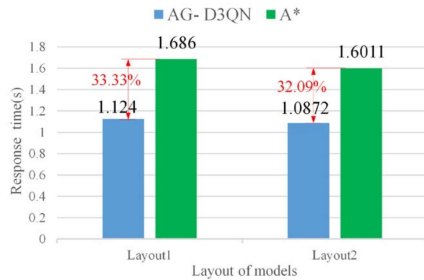


Fig. 7. Average Response Time

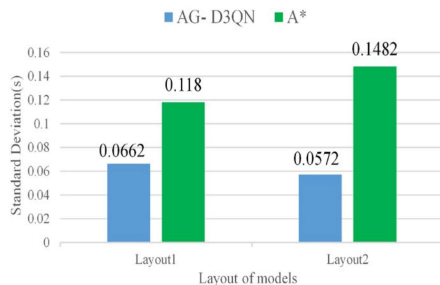


Fig. 8. Standard Deviation

IV. CONCLUSION

In this paper, AG-D3QN algorithm is proposed to optimise the path planning problem in digital workshop. Experiments show that compared with the traditional A* algorithm, AG-D3QN significantly reduces the response time under the premise of guaranteeing the shortest path, and exhibits faster convergence speed than the standard D3QN. However, although AG-D3QN performs well in single AGV path optimisation, its applicability is limited in multi-AGV systems due to the increased environmental uncertainty.

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